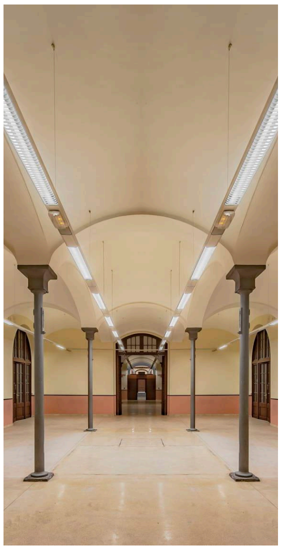
Open-RMF model of the Industrial School (UPC)

First part



Create a model in Open-RMF of the Industrial School (UPC) with the following elements:

- Ground floor only
- Walls and doors only (no lifts, no furniture)
- One fleet with two tinyRobots

The robots should be able to perform patrol tasks around the following locations of the building: Local 1, Local 2, Local 3, Local 4, Local 5, Local 6, Local 7.

You can reuse and/or adapt any file from the Open-RMF Demos repository.

Instructions

1. Create a ROS workspace named “exercise_ws”.
   Create a LibreOffice writer document named “README.odt” in it.
   (You will use this file for documenting your model.)

2. Run the Open-RMF Docker container with the command:
```
rocker --nvidia --x11 --name rmf_library \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  --network host --user --volume `pwd`/exercise_ws:/exercise_ws -- \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest bash
```

3. Download the PNG floorplan of the building from Aula Virtual.

4. Create a building model named “industrial.building.yaml” with `traffic-editor`.
   You should add the walls, doors, lanes, robots, etc.
   Include in the “README.odt” file a description of the created routes and spawned robots.

5. Create a launch file named “industrial.launch.xml” for running the simulation and visualization (you can create additional launch files if needed).
   Add the appropriate instructions for launching the simulation to the “README.odt” file.

6. Add the instructions to the “README.odt” file for running several patrol tasks **in the command line**. Include some figures with snapshots of Gazebo and RViz for each task.

Second part

Create an extended version of the previous model including the new following elements:

- One additional floor (underground level 1 or 2)
- One lift between the floors
- One fleet with two cleanRobots
- One fleet with two deliveryRobots

Add the instructions to the “README.odt” file for running several patrol, cleaning and delivery tasks in the command line. Include some figures with snapshots of Gazebo and RViz for each task.